

News

Office Hours

Instructor - Mon 1-3

TA - Wed 12-2  
Fri 1-3Vectors & Dyadics

- Matlab Tips
- Schedule
- Vector Multiplication

Assessment

Homework 1

Due Fri, 6pm.

Matlab Tips

straight to output vector, skip 'structure'

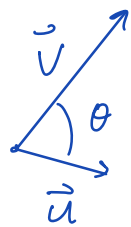
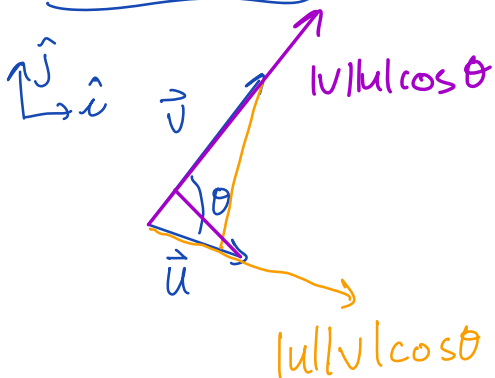
$$[t, X] = \text{ode45}(\text{odefun}, \underbrace{tspan}, \underbrace{X0, options})$$
Can be 2 elements;  $[t0, tf]$ or multiple;  $[t0, t1, t2 \dots tf]$ which can affect integrated time steps,  
and the solution...., 'AbsTol',  $1e^{-5}$ , 'RelTol',  $1e^{-2}$ )

Absolute tolerance, Relative tolerance.

Watch out for floating point accuracy

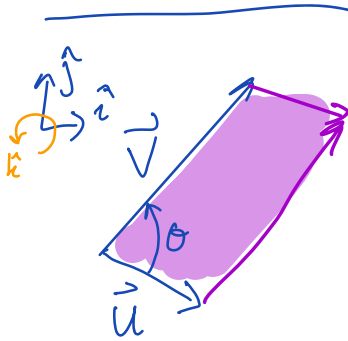
Vector Multiplication

$$\vec{u} = \begin{bmatrix} u_x \\ u_y \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

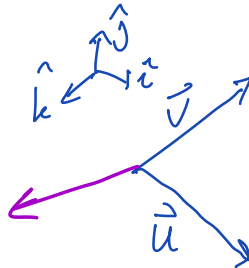
Dot Product  $\vec{u} \cdot \vec{v} = \text{scalar}$ 

$$\vec{u} \cdot \vec{v} = u_x v_x + u_y v_y$$

$$\begin{bmatrix} \hat{i} \cdot \hat{i} = 1 & \hat{j} \cdot \hat{i} = 0 \end{bmatrix}$$

Cross Product  $\vec{u} \times \vec{v} = \text{vector}$ 

$$|\vec{u}| |\vec{v}| \sin \theta \hat{k}$$



$$\vec{u} \times \vec{v} = u_x \hat{i} \times v_y \hat{j} - u_y \hat{j} \times v_x \hat{i}$$

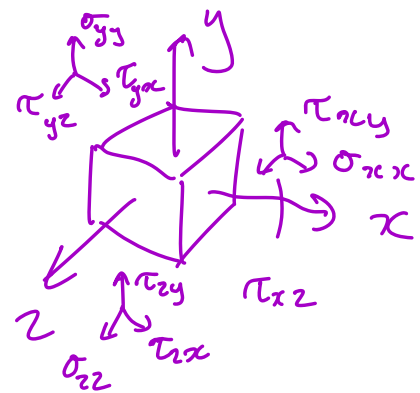
$$= \det \begin{pmatrix} u_x & v_x \\ u_y & v_y \end{pmatrix} \hat{k}$$

$$= \det \begin{pmatrix} \hat{i} & u_x & v_x \\ \hat{j} & u_y & v_y \\ \hat{k} & 0 & 0 \end{pmatrix}$$

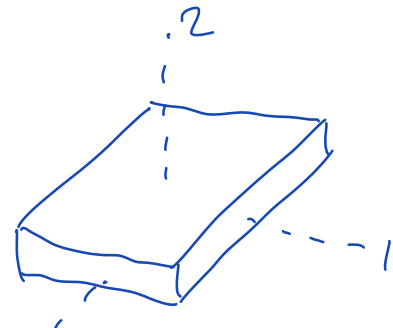
Outer Product  $\vec{u} \otimes \vec{v} = \underline{\text{tensor}}$   $\swarrow$  dyad

$$\vec{u} \otimes \vec{v} = \begin{bmatrix} u_x \\ u_y \end{bmatrix} \begin{bmatrix} v_x & v_y \end{bmatrix} = \begin{bmatrix} u_x v_x & u_x v_y \\ u_y v_x & u_y v_y \end{bmatrix}$$

$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix}$$



$$\underline{\underline{I}} = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix}$$



$$\underline{\underline{R}} = \sum_{i=1}^3 \hat{e}_i' \otimes \hat{e}_i = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

